The untrained baseline for PinoAI's first continued-pretraining test: Filipino, English and a same-question language gap, measured before training

PinoAI research note · RESULT-003 · DEC-005 (ABL-CPT-001), before arm · Taglish edition: RESULT-003-dec005-before-arm.tl.qmd

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Abstract

PinoAI's first training experiment asks whether a short, language-only continued pretraining (CPT) of Gemma 4 E4B makes it better at Filipino without making it worse at English. The success criteria were registered before any training. This note reports the **before** arm: the untrained base model, on the same pinned harness that will score the trained model.

- **Filipino (FilBench, 12,956 multiple-choice items):** 63.36% as the registered macro over 8 groups (63.48% pooled).
- **English guard (3,312 items):** 55.28% pooled (Global-MMLU English 67.72%, ARC-Challenge 56.66%, HellaSwag 39.50%).
- **Same questions, two languages (descriptive):** on 1,140 Global-MMLU questions asked in both English and Tagalog, the model is right in English but wrong in Tagalog on 191 items, and the reverse on 100 (exact McNemar p = 1e-7). That is an 8.0-point language gap that CPT should narrow.
- **Reproducibility:** two independent runs agree on all 2,312 re-scored items, down to the answer log-probabilities.

No claim about CPT is made here. This is the reference point the trained model will be compared against.

1. What is being measured, and why before training

The registered experiment (DEC-005) passes only if three criteria hold together:

- **C1:** the model's bits-per-byte on held-out Filipino text drops by at least 10%;
- **C2:** its Filipino multiple-choice accuracy rises;
- **C3:** its English accuracy falls by no more than 2 points.

Each criterion compares the trained model with the untrained one on the same items. So the untrained numbers have to exist, with the harness frozen, before any training run. That way neither the items nor the scoring code can move after the trained model's scores are seen.

Two choices make the comparison exact:

- **One harness revision.** The evaluation library (lighteval) is pinned to one commit (c2af9e5f) for both arms, and a run whose recorded commit differs is refused.
- **Log-likelihood scoring.** All items are multiple choice scored by comparing the model's probabilities of the answer letters, with no sampling. So the same model gives the same answer every time (Section 5).

2. Filipino: FilBench (C2)

FilBench's Filipino multiple-choice tasks, with the chat template off, the same 12,956 items as an earlier comparison of base models. Item identifiers and prompt texts were checked: all 12,956 are identical to that earlier run, so the harness pin changed no items.

The registered primary metric is the **macro** average over 8 groups, each weighted equally. Two pairs of small tasks are merged into one group each (SIB-200 Tagalog + Cebuano; named-entity recognition Tagalog + Cebuano). Without the merge, those small tasks would add noise to the macro.

Group	Correct / items	Accuracy
Belebele, Cebuano	599 / 900	66.56%
Belebele, Filipino	683 / 900	75.89%
FiReCS	4,329 / 7,340	58.98%
Global-MMLU, Tagalog	681 / 1,140	59.74%
INCLUDE, Tagalog	385 / 500	77.00%
NER, Tagalog + Cebuano	1,333 / 1,628	81.88%
Readability, Cebuano	97 / 350	27.71%
SIB-200, Tagalog + Cebuano	117 / 198	59.09%
Macro over 8 groups (registered primary)		**63.36%**

Pooled over all 12,956 items: 63.48%.

: C2 before arm: FilBench multiple-choice accuracy of the untrained Gemma 4 E4B base, by registered group. {#tbl-c2}

One group sits at chance. Cebuano readability, at 27.71%, is no better than guessing for its format, and earlier base models scored 36.6-39.7% on it. It is one of the 8 equal-weight groups, so its noise enters the macro with weight 1/8. That was registered and is not changed here, but per-group changes will be reported so that a movement in this group can be seen for what it is.

3. English guard (C3)

C3 checks that Filipino training does not cost English ability. It pools 3,312 English multiple-choice items: Global-MMLU English (20 questions per subject, 1,140), the ARC-Challenge test set (1,172), and a seeded random sample of 1,000 HellaSwag validation items (seed 20261001; the sampled set was verified against the registered seed).

Table 1: C3 before arm: English multiple-choice accuracy of the untrained base.

Group	Correct / items	Accuracy
ARC-Challenge (test)	664 / 1,172	56.66%
Global-MMLU, English (20 per subject)	772 / 1,140	67.72%
HellaSwag (seeded 1,000)	395 / 1,000	39.50%
Pooled, 3,312 items (C3 metric)		**55.28%**

A scoring correction, made before this became the reference. In the pinned library, HellaSwag's default scorer generates an answer letter and checks it for an exact match, while ARC and Global-MMLU compare answer-letter probabilities. A first run therefore mixed two scoring rules. HellaSwag was switched to the probability rule used by the other two, so all of C3 is scored one way. The effect was small (393 to 395 of 1,000), since the generated letter is almost always the most probable letter. The change was recorded before any trained model exists, and it applies to both arms.

A caveat on HellaSwag. At 39.50% (chance is 25%), letter-format HellaSwag is hard for a 4B base model, and it partly measures whether the model follows the answer-letter format. C3′s registered guard is the pooled accuracy, and HellaSwag is 1,000 of its 3,312 items. A change confined to HellaSwag will be reported separately.

4. The same questions in two languages (descriptive)

Global-MMLU asks the same questions in many languages. The C2 run includes 1,140 of them in Tagalog and the C3 run the same 1,140 in English, so each question can be compared with itself. The pairing was checked: the correct answer is the same option on all 1,140 pairs.

	Right in Tagalog	Wrong in Tagalog
Right in English	581	191
Wrong in English	100	268

English 772 / 1,140 (67.72%), Tagalog 681 / 1,140 (59.74%); exact McNemar p = 1e-7.

: The same 1,140 Global-MMLU questions in English and in Tagalog, untrained base. Discordant pairs: right in one language only. {#tbl-gap}

The untrained model answers in English what it misses in Tagalog almost twice as often as the reverse: 191 against 100 (exact McNemar p = 1e-7, an 8.0-point gap). Because each question is compared with itself, this gap does not depend on any one benchmark's quirks. Filipino CPT should narrow it. The measure was not registered as a criterion, so it will be reported before and after as a **descriptive** result only.

5. Reproducibility

The English guard was run twice, in two separate cloud containers, before the scoring correction in Section 3. On the 2,312 ARC and Global-MMLU items, which both runs scored the same way, every item's outcome was identical, and so was every answer-letter log-probability to four decimal places. A difference between the before and after arms will therefore be the model's, not run-to-run noise.

6. A code-path check of C1 (not a baseline)

C1′s baseline needs the held-out slices of the full training corpus, which is not built yet. To test the code that will compute it, the bits-per-byte scorer was run on a small pilot build (270 held-out web documents and 13 out-of- distribution literary works, all Tagalog). **These numbers are a code-path check, not the C1 baseline:**

- held-out web Tagalog: 0.885 bits per byte;
- out-of-distribution literary Tagalog: 1.421 bits per byte.

The literary slice is harder for the model, as expected, since its older, literary Tagalog is further from typical web text. The scorer reuses the training code's own evaluation path, and a unit test confirms that its per-document numbers add up to the trainer's aggregate.

7. What this note does and does not show

- It fixes the reference numbers for DEC-005′s C2 and C3, on a harness that cannot drift between arms.
- It does not test the CPT hypothesis. That needs the trained model's after arm.
- The C1 baseline is still to come, from the full corpus build.
- The language gap in Section 4 is descriptive, not a registered criterion.

Reproducibility

- Before-arm result file: `experiments/results/DEC-005-before-arm.txt`, from `scripts/eval/score_dec005_before.py`. Both files are hash-registered, and the after arm uses the identical script.
- Runs: `data/eval/runs/google-gemma-4-e4b/dec005-c2/` and `dec005-c3/`, run metadata recording the pinned harness commit, the GPU and the sampling seed.
- Tables in this note: `experiments/reports/_result003.py`, which reads the result file and asserts every quoted number.

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